Cross-Domain Self-Supervised Pretraining for Robust Driving Perception

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Abstract

Driving perception models degrade under conditions they were not trained on, such as fog, night or a new city. We ask whether self-supervised pretraining on *more diverse* unlabeled driving footage improves robustness to such shifts, and whether this is worth more than additional labeled data from a single domain. We pretrain a ViT-S/16 with masked autoencoding (MAE; MoCo v3 as an ablation) on pools of driving frames of *identical size and compute* that differ only in diversity (single-condition BDD100K, condition-balanced BDD100K, and BDD100K+nuScenes+Waymo). Each model is fine-tuned on one labeled source domain (BDD100K) and evaluated on shifted targets (BDD night/rain/snow, nuImages, Waymo, Cityscapes, ACDC), reporting both raw and *effective* robustness. [TODO: One sentence with the headline quantitative result once experiments finish.]

1 Introduction

Autonomous-driving perception is deployed across cities, weather and lighting that no labeled training set covers exhaustively. Labels are expensive, whereas unlabeled driving video is abundant. Self-supervised learning (SSL) can exploit this, but a basic question is open: when the unlabeled data budget is fixed, *what composition of footage* yields the most robust representation? Comparing a large diverse corpus with a small narrow one confounds diversity with scale, so we control for dataset size and pretraining compute and vary only the diversity of the pool.

Contributions. (i) A controlled protocol with iso-size pools and iso-step pretraining that isolates diversity from scale; (ii) a multi-dataset, multi-condition evaluation of segmentation and detection with effective-robustness analysis; (iii) a comparison of the benefit of diverse pretraining against extra single-domain labels (an "exchange rate"); (iv) an open codebase reproducing all experiments.

2 Literature Review

Self-supervised vision transformers. MAE [6] reconstructs masked patches and is memory-efficient because the encoder sees only visible tokens; MoCo v3 [2] learns invariances via contrastive views. DINOv2 [7] shows that curating data for diversity matters for the quality of learned features.

Robustness under distribution shift. Taori et al. [11] separate genuine robustness from improvements that merely follow in-domain accuracy through *effective robustness*, which we adopt.

Driving datasets and domain generalization. BDD100K [12], nuScenes [1], Waymo [10], Cityscapes [4] and ACDC [9] cover different geographies, sensors and conditions. Domain-generalization methods for segmentation such as RobustNet [3] modify the training objective; we instead study the *data* used for pretraining. [TODO: Add driving-specific SSL work.]

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3 Problem Statement

Let $\mathcal{U}$ be a union of unlabeled driving datasets and $P \subset \mathcal{U}$ a pool with $|P| = n$. A backbone f_P is pretrained on P for a fixed number of optimizer steps, then fine-tuned on labeled data from a single source domain $\mathcal{D}_s$. For each target domain $\mathcal{D}_t$ we measure the metric $m_t(f_P)$ (mIoU or AP). We test whether, for fixed n and compute, pools of higher diversity give higher m_t on shifted targets, in particular beyond what the in-domain score m_s predicts. Hypotheses H1–H5 are: (H1) diverse pools improve OOD performance; (H2) this holds for targets whose dataset is absent from the pool; (H3) a diverse pool beats a $3\times$ larger single-source pool; (H4) diverse pretraining is worth $k\times$ more single-domain labels; (H5) the effect is not specific to MAE.

4 Proposed Method

Pools. Frames from BDD100K, nuScenes and Waymo are resized to a 256 px short side. `build_pool` returns exactly n frames (it raises otherwise), using source weights, filters and water-filled balancing over time of day and weather, and excludes all evaluation images to prevent leakage.

Pretraining. ViT-S/16 trained with MAE (75% masking, normalized-pixel loss) or MoCo v3 with temporal positives. Length is fixed in *steps*, identical across pools.

Fine-tuning. Semantic FPN (and a linear probe) on Cityscapes-19 classes, and Faster R-CNN [8] on a ViT pyramid for harmonized *vehicle/pedestrian* detection, with layer-wise learning-rate decay. The labeled subset depends only on file, filter, fraction and subset seed.

Metrics. mIoU/AP per domain; mean and worst OOD; effective robustness as the residual of OOD from a logit-linear fit of OOD on ID over baseline (random/ImageNet) models [11]; pool diversity via metadata entropy and the Vendi score [5]. No checkpoint or hyper-parameter is chosen on OOD targets.

5 Experiments and Results

Setup. Datasets, pools, recipes and compute as in the repository (`docs/RESEARCH_PLAN.md`). Experiments: E1 diversity, E2 leave-one-dataset-out, E3 scale vs. diversity, E4 label scaling, E5 method. [TODO: Figures: ID vs OOD scatter (effective robustness), diversity vs OOD, label scaling, MAE reconstructions (`scripts/visualize_mae.py`), pool composition (`scripts/describe_pools.py`), qualitative fog/night (`scripts/demo.py`). Include with `\includegraphics`.]

Table 1: Per-domain performance (mean±std over seeds). [TODO: Fill from `outputs/e1_diversity/*_summary.csv`.]

Init	ID	mean OOD	worst OOD
Random	[TODO: –]	[TODO: –]	[TODO: –]
ImageNet	[TODO: –]	[TODO: –]	[TODO: –]
MAE narrow	[TODO: –]	[TODO: –]	[TODO: –]
MAE diverse	[TODO: –]	[TODO: –]	[TODO: –]

6 Conclusions

[TODO: Summarize findings against H1–H5, limitations (18K-frame budget, single laptop GPU, one pretraining seed, small target slices), and future work (larger pools, temporal SSL, ConvNet backbones).]

References

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